CoMET: Tracking the Shared Pose Error of Roadside Messages for Asynchronous Vehicle–Infrastructure Cooperative 3D Detection

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Abstract

A roadside unit can see the car that a truck hides from the ego vehicle, but in deployment its message arrives late and can be placed only with poses that are available online. We study vehicle–infrastructure cooperative 3D detection on V2X-Seq under a strict arrival protocol that admits only messages captured before the decision time and uses no ground truth, future frames or per-pair calibration offsets at inference. In this setting most of the value of cooperation is lost to geometry rather than to latency: late fusion with online poses improves AP@0.7 over the ego detector by only 1.0 point and intermediate fusion by at most 0.8, whereas the same late fusion placed with the dataset's annotation-assisted offsets would improve it by 7.4. The loss is a slowly drifting translation shared by all objects of a message. The objects seen by both agents reveal it, but 38% of messages contain no such object.

We propose CoMET (Common-Mode Error Tracking), which treats this translation as a latent state of the message stream: every ego detection is an anchor candidate; one observation per message is formed by mutual-nearest-neighbour Hungarian matching; a causal scalar Kalman filter tracks the offset per sender; and the estimate corrects every remote box, including those of objects the ego vehicle cannot see. A lock-acquisition step recovers offsets larger than the association gate from a vote over recent displacements. The estimator has no learned parameters, and messages carry only boxes, scores and capture times; a 1.3K-parameter readout head, trained once and frozen, re-ranks the merged candidates.

CoMET reaches 66.63 AP@0.7, 3.45 points above the same late fusion with online poses and 3.67 above the best intermediate fusion. It improves all five detectors we test, from three architectures, by 2.2–3.5 points over online poses and by 0.3–0.6 over estimating the offset from the current message alone. Nine variants of three per-frame registration modules, with and without temporal smoothing, remain 2.3–3.1 points behind. We also find that the benchmark's reference offset is an exact-zero placeholder on 28% of validation messages and comes from a bounded discrete search that saturates on 45% of the remaining ones; because the roadside-only ground truth is built from this offset, the official evaluation inherits both artefacts. We therefore report results on label-defined populations and under two auxiliary references; CoMET improves on every population under both. Under injected pose errors of 3–5 m, CoMET with lock acquisition retains 97.6–99.5% of its accuracy, whereas tracking alone falls nearly to the level of online poses. Finally, retraining an intermediate-fusion network with CoMET poses and cleaned labels improves over retraining with online poses on all three architectures we test, significantly on two; combining its boxes with those of CoMET late fusion gives the best result overall, 67.85 AP@0.7 and 82.62 AP@0.5.

1 Introduction

An ego vehicle approaching an intersection cannot see the car hidden behind a truck; a LiDAR mounted on a pole above the intersection can. Telling the vehicle where that car is, early enough to plan around it, is the promise of vehicle–infrastructure (V2I) cooperative perception [27, 29]. Most

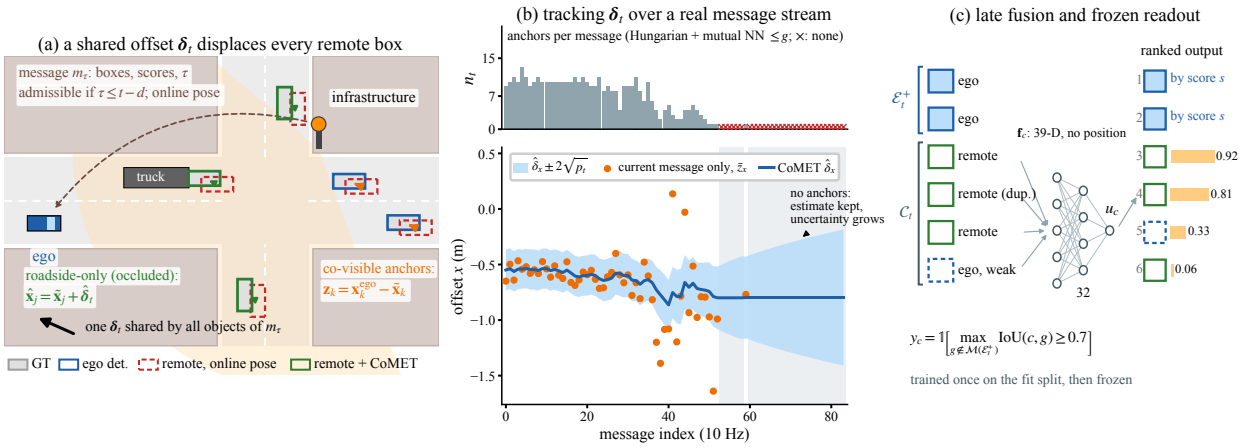


Figure 1: CoMET at a glance. (a) A roadside message is admissible only if it was captured at least d ms before the decision time t. Placed with online poses, all remote boxes (red, dashed) are displaced from the ground truth (gray) by the same offset $\boldsymbol{\delta}_t$ (exaggerated here). Objects seen by both agents yield anchor residuals $\mathbf{z}_k$ between ego boxes (blue) and remote boxes that measure $\boldsymbol{\delta}_t$; the tracked estimate moves every remote box, including the roadside-only objects hidden behind the truck (green). (b) A real validation stream (sequence 0071, $d = 0$). Anchors per message come from Hungarian matching of all ego detections with a mutual-nearest-neighbour gate. The estimate from the current message alone (orange) is noisy and absent when there are no anchors (gray bands); the CoMET estimate (blue) is stable, is kept through anchor-free messages and its uncertainty grows until anchors return. (c) Confident ego detections form a protected prefix $\mathcal{E}_t^+$; corrected remote boxes and weak ego boxes $\mathcal{C}_t$ are ranked by a 1.3K-parameter readout head on 39 receiver-side features without absolute position, trained once and frozen. Only boxes, scores and capture times are transmitted.

methods fuse the two views at the feature level [2, 5, 11, 16, 19, 20] or at the box level [27], and most reported gains are measured with views that are synchronised and registered with carefully calibrated poses.

A deployed receiver has neither. The roadside message was captured before the ego frame and arrives after a communication delay, and it can only be placed with the poses that are available online: the vehicle's localisation estimate and the roadside calibration. We evaluate V2X-Seq [29] under a *strict arrival protocol* that admits only messages captured at least d ms before the decision time ($d \in \{0, 100, 200, 400\}$) and that never uses ground truth, future frames or the dataset's per-pair calibration offsets at inference. Under this protocol, cooperation as currently practised adds little (Fig. 2): late fusion with online poses raises AP@0.7 over the ego detector by 1.0 point, and three intermediate-fusion models change it by -1.0 to $+0.8$. Yet the value is there: the same late fusion placed with the dataset's annotation-assisted offsets would raise AP@0.7 by 7.4 points. The decisive limitation is therefore geometric, not temporal.

The error is shared, persistent and sparsely observed. V2X-Seq provides one planar translation per frame pair, and applying it recovers most of the lost accuracy; a single translation shared by all objects of a message—a *common-mode* error—therefore explains most of the misplacement. This error is observable without any label: an object seen by both agents (an *anchor*) yields a residual between the ego box and the projected roadside box that measures the shared translation, and the same correction then applies to the roadside-only objects that motivate cooperation in the first place. A single message, however, carries little evidence. On the validation sequences a message holds 2.96 anchors on average, and 38% of messages hold none; estimates from the current message

alone are therefore often impossible and, when possible, noisy (a radial error of 0.48 m against a smoothed reference at 0 ms and 0.61 m at 400 ms). At the same time the error drifts slowly within a sequence (Fig. 3). The misalignment is a property of the *message stream*, not of any individual frame.

Tracking instead of registering. Latency-aware methods [9, 15, 17, 24, 28] compensate motion between capture and use and assume that the relative pose is right. Pose-robust methods [6, 10, 13, 14] re-register each frame from its own correspondences and are mostly evaluated under independent synthetic noise. Neither treats the partner's error as persistent: the first leaves it uncorrected, and the second re-estimates it from scratch in every frame, so no estimate is available when correspondences are missing. We propose CoMET (Common-Mode Error Tracking), which treats the partner's current systematic error as a state to be estimated together with its uncertainty. CoMET uses every ego detection as an anchor candidate, forms one robust observation per message with mutual-nearest-neighbour Hungarian matching, tracks the shared offset with a causal Kalman filter per sender, carries the estimate through messages without anchors, and transfers it to every remote box (Fig. 1). When the offset is larger than the association gate, a lock-acquisition step recovers it from a vote over the displacements of recent messages. A small frozen readout head then re-ranks the corrected boxes together with the low-score ego detections. Because process and measurement noise are isotropic, the filter reduces exactly to a scalar recursion with a two-coordinate state and one variance per stream; it has no learned parameters and needs only the boxes that a late-fusion message already carries. Its two limiting cases are the two classical answers to pose error, per-frame registration and fixed calibration, and on real data it outperforms both.

The benchmark's reference is itself incomplete. Evaluating such a method requires care, because the reference offsets of V2X-Seq are not what they appear to be. We audit them against the raw annotations and find that (i) the offset is *exactly zero* on 28.0% of validation messages, nearly all of which contain no co-labelled object, so zero is a placeholder rather than a measurement; (ii) the roadside-only ground truth is the public projection plus this offset (residual $\approx 4 \times 10^{-7}$ m), so on placeholder frames the ground truth itself is uncorrected; and (iii) the non-zero offsets come from a bounded discrete search with 26 values per axis whose bound is reached on 44.6% of non-zero validation messages, where the co-labelled objects are typically displaced beyond the bound (Fig. 4). An estimator that is right on a placeholder frame is therefore penalised by the official evaluation, and the dataset offset is a label-consistent reference rather than an oracle. We report the official evaluation as the primary result and complement it with label-defined populations and two auxiliary references; under both auxiliary references CoMET improves on every population.

Results. With the boxes of the released TraF-Align detector, CoMET reaches 66.63 AP@0.7 averaged over the four latency budgets, 3.45 points above late fusion with online poses and 3.67 above the best intermediate fusion, while keeping the advantage at every budget. It improves the late fusion of all five detectors we test by 2.2–3.5 points over online poses and by 0.3–0.6 over estimating the offset from the current message alone. Nine variants of three per-frame registration modules, with and without temporal smoothing, remain 2.3–3.1 points behind, and as the pose input of CoAlign, CoMET beats CoAlign's own pose graph by 1.1 points. With known pose errors of 3–5 m injected into the roadside messages, CoMET with lock acquisition keeps 97.6–99.5% of its accuracy for all five detectors, and its corrected pose raises the accuracy of retrained intermediate fusion under the same errors by 2.2–4.4 points. When intermediate fusion is retrained, using CoMET poses and cleaned labels is better than retraining with online poses on all three architectures we

test (significantly on two). Combining the boxes of the retrained network with those of CoMET late fusion reaches 67.85 AP@0.7 and 82.62 AP@0.5, better than either alone on both metrics.

Contributions.

- We show that under strict arrival and online poses the main obstacle to real V2I cooperative detection is a slowly drifting common-mode translation, which co-visible objects reveal but which no single message estimates reliably.
- We propose CoMET, a training-free causal tracker of this translation with cross-object transfer, which interpolates between per-frame registration and fixed calibration, re-acquires offsets beyond its association gate, and costs a single small assignment per message.
- We audit the reference offsets of V2X-Seq, identify a zero placeholder and a saturated bounded search, explain how they enter the official ground truth, and propose an evaluation that separates label artefacts from method effects.
- We provide an extensive evaluation with paired sequence-level confidence intervals, hyper-parameters selected on a separate calibration split only, every reported number re-derived by an independent implementation, a controlled study with injected pose errors on real data, and a retraining study across three architectures together with a box-level combination of late and intermediate fusion.

2 Related Work

Cooperative 3D detection. Cooperative perception exchanges raw point clouds (early fusion), intermediate features, or detected boxes (late fusion). Intermediate fusion dominates recent work: F-Cooper [2] max-pools voxel features, V2VNet [16] aggregates them with a graph network, DiscoNet [11] distils an early-fusion teacher, V2X-ViT [20] and CoBEVT [19] attend across agents, and When2com [12] and Where2comm [5] reduce bandwidth through selective communication. Benchmarks include OPV2V [21] in simulation, DAIR-V2X [27] and V2X-Seq [29] for real V2I, and V2V4Real [22] for real V2V. Late fusion suits scarce bandwidth and heterogeneous detectors but is generally considered weaker [27]. We find that, once the shared geometric error is tracked, box-level fusion outperforms the intermediate-fusion models we test at the strict IoU threshold, and that the same pose estimate also helps when intermediate fusion is retrained.

Asynchrony and latency. A partner's message is captured at another time and arrives with delay. SyncNet [9] compensates latency with a feature-level temporal model, CoBEVFlow [17] transports BEV features along estimated object flow, FFNet [28] transmits feature flow so that the receiver can predict the roadside feature at the ego time, How2comm [24] couples flow-guided compensation with communication selection, TraF-Align [15] aligns features along predicted trajectories, and CoDynTrust [23] modulates features with a dynamic trust estimate. These methods model the *motion* between capture and use and assume a correct relative pose. On real V2I data with online poses we find the larger error to be geometric and persistent; CoMET addresses it at the box level and is complementary to motion compensation.

Pose errors in cooperative perception. V2X-ViT [20] learns robustness to injected pose noise; CoAlign [13] optimises an agent–object pose graph built from predicted boxes and their uncertainties in every frame; FreeAlign [10] matches salient-object graphs to recover the relative pose without

external localisation; RoCo [6] alternates object matching and pose adjustment; CBM [14] matches objects through their local context; LET-VIC [25] compensates calibration errors inside a BEV cross-attention module for end-to-end V2I tracking. Apart from the learned LET-VIC module, these methods estimate the relative pose *from the current frame*, and they are mostly evaluated with independent Gaussian perturbations of synchronous poses. CoMET instead exploits the temporal persistence of the error, accumulates evidence over the message stream with a causal filter, and transfers the estimate to objects without any correspondence in the current frame. We compare with FreeAlign, RoCo and CBM, with and without temporal smoothing of their outputs, under an identical readout, and we feed CoMET poses to CoAlign.

Temporal filtering and online calibration. Kalman filtering [7] underlies tracking-by-detection [18, 26], where each object carries its own state. Iterative registration [1] recovers a rigid transform from dense correspondences between a single pair of scans. CoMET filters a single *shared* state per partner, the offset common to all of its objects, which is closer in spirit to online extrinsic calibration than to object tracking, and it estimates this state from sparse, noisy correspondences accumulated over time. It needs neither raw point clouds nor calibration targets: everyday traffic seen by both agents serves as the calibration target.

Reference quality of cooperative benchmarks. Cooperative datasets provide calibration and pose refinements that are produced offline, and models are commonly trained and evaluated with them [22, 27, 29]. We are not aware of an audit of how such refinements are generated and how they enter the cooperative ground truth. Our analysis of V2X-Seq shows that both questions matter for evaluating any method that corrects geometry online.

3 Problem: Where the Value of Cooperation Is Lost

3.1 Setting

An ego vehicle e and a roadside unit i each run a LiDAR detector. At decision time t the ego detector outputs boxes $\mathcal{B}_e(t) = \{(\mathbf{b}_k, s_k)\}$, and the roadside unit broadcasts messages $m_\tau = (\mathcal{B}_i(\tau), \tau)$ with its detections at capture time τ. Under the *strict arrival protocol* the receiver may use only messages with $\tau \leq t - d$ for a latency budget d; frames without an admissible message are still evaluated. Remote boxes are mapped into the ego frame with the poses available online—the vehicle localisation $\mathbf{T}_{w \leftarrow e}(t)$ and the roadside calibration $\mathbf{T}_{w \leftarrow i}$—so that $\mathbf{T}_{e \leftarrow i}(t) = \mathbf{T}_{w \leftarrow e}(t)^{-1} \mathbf{T}_{w \leftarrow i}$. No ground truth, future frame or per-pair calibration offset is used at inference. The task is to detect vehicles in the ego frame at time t.

3.2 Geometry, not latency, dominates

Figure 2 shows AP@0.7 on the 21 validation sequences against the latency budget (exact numbers in Table 2). Three observations frame the rest of the paper. *(i) Cooperation barely helps as practised.* Late fusion of the released TraF-Align [15] boxes with online poses improves AP@0.7 over the ego detector by 1.0 point, to 63.18 on average, and the official late-fusion baselines and three intermediate-fusion models stay between 61.1 and 63.0. *(ii) The value is recoverable.* Placing the same remote boxes with the dataset's annotation-assisted offset, one planar translation per frame, gives 69.52 AP@0.7: one translation per message adds 6.3 points over online poses and turns a 1.0-point gain of cooperation into a 7.4-point gain. *(iii) Latency costs little in comparison.* With

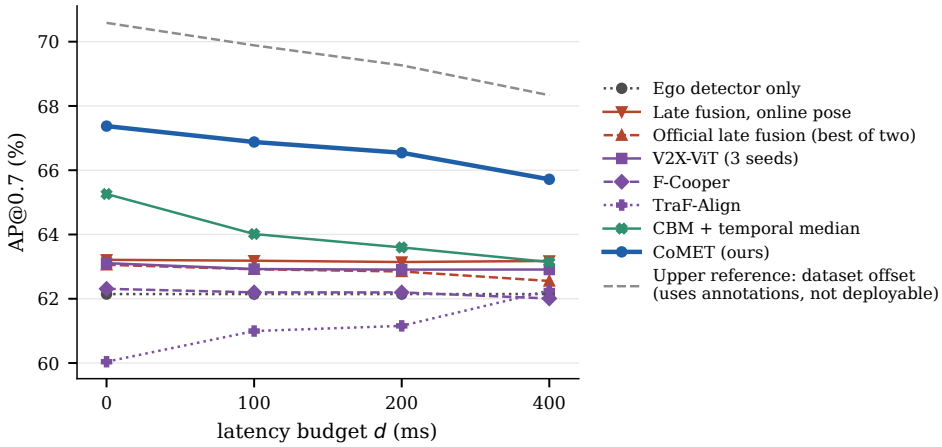


Figure 2: AP@0.7 against the latency budget on V2X-Seq validation. Intermediate fusion and late fusion with online poses stay close to the ego detector. CoMET recovers over half of the gap to the upper reference that places the boxes with the dataset's annotation-assisted offset (dashed gray, not deployable). Values are means over three readout heads or three detector seeds; numbers in Table 2.

good geometry, AP@0.7 drops by 2.3 points from 0 to 400 ms with the dataset offset (70.59 to 68.33) and by 1.7 points with CoMET (67.37 to 65.72), well below the geometric gap.

3.3 A shared, persistent, sparsely observed error

We model the centre of remote object j after mapping to the ego frame as

$$\tilde{\mathbf{x}}_j = \mathbf{x}_j - \boldsymbol{\delta}_t + \boldsymbol{\epsilon}_j, \tag{1}$$

where $\mathbf{x}_j$ is its true centre, $\boldsymbol{\delta}_t$ a planar correction shared by all objects of the message (the common mode) and $\boldsymbol{\epsilon}_j$ an object-specific error from detection noise and residual motion. Such a shared term arises when the dominant error lies in the transform that maps the roadside frame into the ego frame—a residual calibration or localisation bias—which displaces every object of a message alike, whereas detection noise and uncompensated motion act on each object separately. Three properties make this model useful.

Observable. For an object seen by both agents, the residual $\mathbf{z} = \mathbf{x}^{\text{ego}} - \tilde{\mathbf{x}}$ between the ego box and the projected remote box equals $\boldsymbol{\delta}_t$ plus noise, with no label involved. The same $\boldsymbol{\delta}_t$ then corrects the remote objects that the ego vehicle cannot see.

Sparse per message. Using all ego detections and mutual-nearest-neighbour matching within 2.5 m (Section 4), a validation message contains 2.96 anchors on average at $d = 0$ (2.73 at 400 ms), and 38.2% of messages contain none; messages that do contain anchors hold 4.79 of them. Estimating $\boldsymbol{\delta}_t$ from the current message alone is therefore often impossible and, when possible, noisy: against a smoothed reference built from the dataset offsets (only used for analysis), the per-message estimate has a radial RMSE of 0.48 m at $d = 0$ and 0.61 m at $d = 400$ ms on messages with anchors, and misses by more than 1 m on 4% and 9% of them; moving anchors increasingly mix the pose error with uncompensated motion as the message ages. The tracked estimate of CoMET has 0.41 m RMSE at both budgets and misses by more than 1 m on 2%.

Persistent. Within a sequence the offset changes slowly compared with the 10 Hz message rate (Fig. 3). The common-mode error should therefore be *tracked* over the message stream rather than re-estimated in every frame.

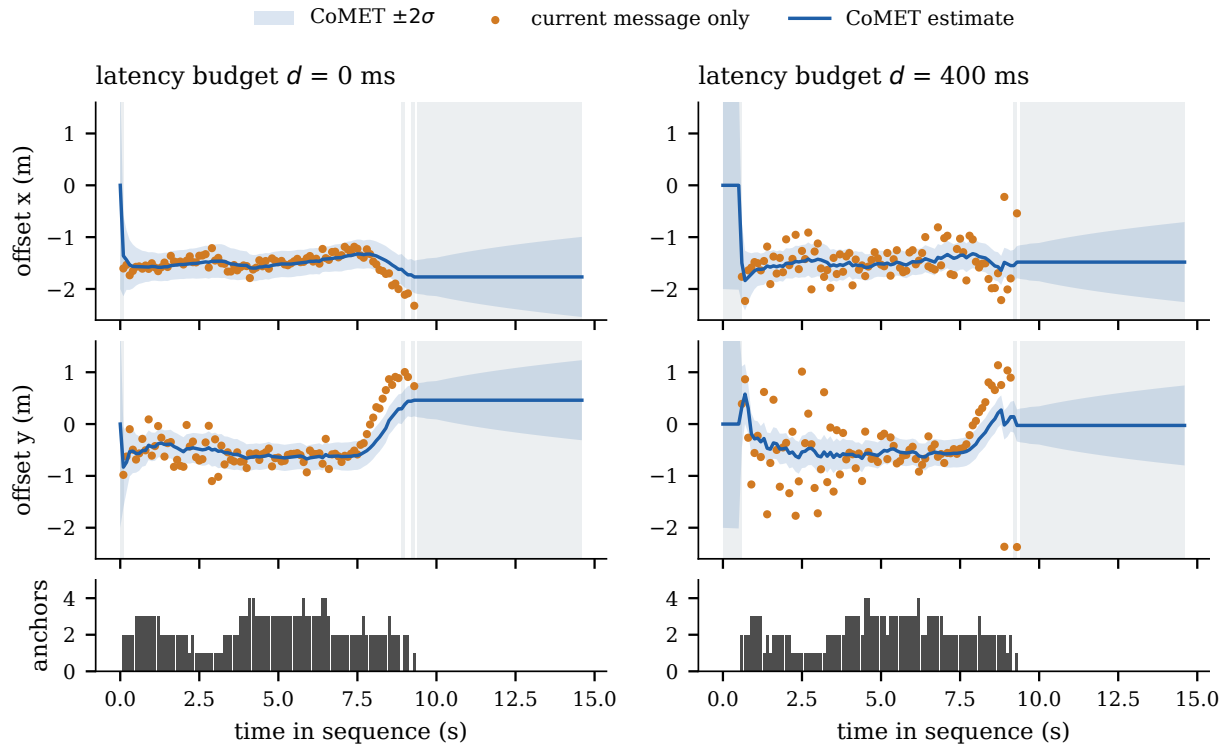


Figure 3: One validation stream (sequence 0007) at two latency budgets. Orange: offset estimated from the anchors of the current message only (shown only when the message has anchors). Blue: the CoMET estimate with its $\pm 2\sigma$ band; gray bands mark messages without anchors. Bottom: number of anchors per message. The per-message estimate is noisy and disappears when anchors are absent (after about 9 s); the tracked estimate is stable, follows the change in y around 8 s, and keeps the last estimate with growing uncertainty when anchors vanish. At 400 ms the per-message estimate is noticeably noisier.

3.4 The benchmark's reference offset

V2X-Seq provides a per-pair `system_error_offset`, a planar translation estimated offline for each vehicle–roadside frame pair. Prior work trains and evaluates with these offsets as the calibrated relative pose. Because they also define what "correct geometry" means in Section 3.2, we audit them directly against the raw annotations of all 67 training and validation sequences (Table 1, Fig. 4; details in Section B).

A placeholder. The offset is exactly zero on 21.8–28.0% of the messages of each split. Outside the calibration split, no zero message contains a co-labelled object (an object annotated by both agents), and only 10–16 of the messages without a co-labelled object have a non-zero offset. Zero is therefore a placeholder for "could not be estimated", not a measurement of zero error. The calibration split contains 35 zero messages that do have a co-labelled object; there the roadside and vehicle labels of that object differ by 1.22 m (median), so the placeholder is not a statement that the error vanished.

The offset is inside the ground truth. For roadside-only objects, the cooperative ground truth equals the public projection of the roadside label plus the offset, with a residual of about 4×10^{-7} m in both the zero and the non-zero case. On placeholder messages the ground truth of roadside-only

Table 1: Census of the dataset offset per split. Messages with an offset of exactly $(0, 0)$; among them, those that still contain a co-labelled object; non-zero messages without a co-labelled object; and non-zero messages with at least one axis on the search bound (-1.1 or $+0.66$ m), as a fraction of non-zero messages.

Split	Messages	Exactly zero	Zero with co-labelled obj.	Non-zero without co-labelled obj.	Non-zero on a bound
fit (24 seq.)	3762	1001 (26.6%)	0	16	1379 (49.9%)
calibration (8)	1247	319 (25.6%)	35	1	721 (77.7%)
held-out (14)	2436	530 (21.8%)	0	10	810 (42.5%)
validation (21)	3316	928 (28.0%)	0	11	1066 (44.6%)

objects is therefore the *uncorrected* projection. A method that estimates the true shift on such a message moves its boxes away from this ground truth and is penalised; a method that does nothing is rewarded.

A bounded discrete search. The non-zero offsets take only 26 distinct values per axis including zero, in five groups with a step of 0.04 m, bounded by -1.1 and $+0.66$ m (Fig. 4a,b). At least one axis sits on a bound for 44.6% of the non-zero validation messages (77.7% in the calibration split). On the bound the co-labelled objects of the same frame are typically displaced further: on the $x = -1.1$ bound (2,413 frames with at least three co-labelled objects) their median displacement is -1.37 m and 91% of the frames lie beyond the bound; on the $y = +0.66$ bound the median is $+0.93$ m. Across frames with at least three co-labelled objects, displacements larger than 1.5 m occur in 19.6% of frames but official offsets in only 1.1%. We infer the search structure from the values alone; the generating procedure is not documented.

Consequences for evaluation. The dataset offset is a label-consistent reference, not an oracle: it is zero where it could not be estimated and clipped where the error is large, and the ground truth inherits both properties. We keep the official ground truth for all primary results and add two complementary readings (Section 5): results split into label-defined populations (all, exactly zero, with co-labelled objects), and two auxiliary references that re-register roadside-only ground truth on placeholder frames (by interpolating the non-zero offsets of the same sequence) or on all frames with at least three co-labelled objects (by their median displacement). Neither auxiliary reference is physical truth; they show which conclusions depend on the label artefacts.

4 CoMET: Common-Mode Error Tracking

CoMET runs on the receiver and treats both detectors as black boxes (Fig. 1, Algorithm 1). It builds remote candidates at the decision time (Section 4.1), forms one anchor observation per message (Section 4.2), tracks the common-mode offset with a causal filter (Section 4.3), re-acquires the offset when it leaves the association gate (Section 4.4), transfers it to every remote box and ranks the merged candidates (Section 4.5). Section 4.6 describes an optional second tier that retrains intermediate fusion with CoMET poses, and a box-level combination of the two tiers.

4.1 Remote candidates at the decision time

A message m_τ carries boxes $\mathbf{b}_j = (\mathbf{x}_j, l_j, w_j, h_j, \theta_j, s_j)$ in the roadside frame. After per-message non-maximum suppression, the receiver associates m_τ with the previous admissible message $m_{\tau'}$ in

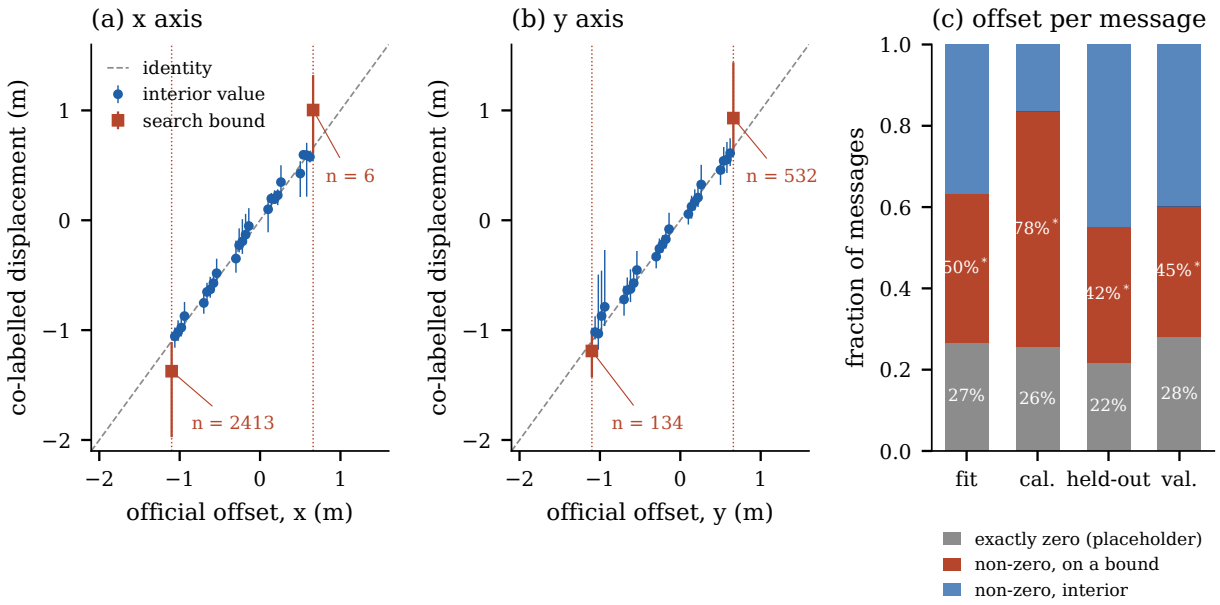


Figure 4: The dataset offset is a bounded discrete search with a zero placeholder. (a,b) For each value the offset takes on an axis, the median and 10–90% range of the displacement of co-labelled objects in the same frame (frames with at least three co-labelled objects, all splits). Interior values follow the diagonal; on the bounds (red) the objects are displaced further than the offset. (c) Per split, the share of messages with an exactly zero offset (gray, percentage of all messages), a non-zero offset on a bound (red; the percentage marked * is relative to non-zero messages), and an interior non-zero offset (blue).

world coordinates and propagates every associated box to the decision time,

$$\tilde{\mathbf{x}}_j = \mathbf{T}_{e\leftarrow i}(t)\big(\mathbf{x}_j^{\tau} + \mathbf{v}_j\,(t-\tau)\big), \qquad \mathbf{v}_j = \frac{\mathbf{x}_j^{\tau} - \mathbf{x}_{a(j)}^{\tau'}}{\tau - \tau'}, \tag{2}$$

where $a(j)$ is the associated box of the previous message and unassociated boxes keep $\mathbf{v}_j = \mathbf{0}$. In Eq. (2) we use the heading-constrained velocity $(\mathbf{h}_j^{\top}\mathbf{v}_j)\,\mathbf{h}_j$, with $\mathbf{h}_j = (\cos\theta_j, \sin\theta_j)^{\top}$, which suppresses lateral velocity noise. Only past messages are used.

4.2 Anchors and the per-message observation

Every ego detection is an anchor candidate; we apply no score threshold (Section 6.4 shows that thresholds only remove evidence). Let $\mathbf{X}^{\text{ego}}$ be the ego box centres and $\tilde{\mathbf{X}}$ the propagated remote centres of the current message. We solve the assignment $\pi^{\star} = \arg\min_{\pi} \sum_k \|\mathbf{x}_k^{\text{ego}} - \tilde{\mathbf{x}}_{\pi(k)}\|$ with the Hungarian algorithm [8] and keep a pair only if it is within a gate g and the two boxes are each other's nearest neighbour:

$$\mathcal{A}_t = \{(k,j) :\ j = \pi^{\star}(k),\ j = \mathrm{NN}(k),\ k = \mathrm{NN}(j),\ \|\mathbf{x}_k^{\text{ego}} - \tilde{\mathbf{x}}_j\| \le g\}. \tag{3}$$

By Eq. (1), an anchor residual observes the common mode directly,

$$\mathbf{z}_{kj} = \mathbf{x}_k^{\text{ego}} - \tilde{\mathbf{x}}_j = \boldsymbol{\delta}_t + \boldsymbol{\eta}_{kj}, \qquad \boldsymbol{\eta}_{kj} \sim \mathcal{N}(\mathbf{0}, r^2\mathbf{I}), \tag{4}$$

where $\boldsymbol{\eta}_{kj}$ lumps the ego and remote object-specific errors. For $n_t = |\mathcal{A}_t|$ anchors, the mean residual $\bar{\mathbf{z}}_t$ is a sufficient statistic, $\bar{\mathbf{z}}_t \sim \mathcal{N}(\boldsymbol{\delta}_t, \frac{r^2}{n_t}\mathbf{I})$. Matching is done on the propagated centres *before*

any correction, so a large current error cannot be hidden by a wrong prediction; using the filter prediction to gate the association ("prediction gating") is worse in our experiments (Table 5). The mutual-nearest-neighbour test removes most ambiguous pairs in dense traffic.

4.3 Tracking the common mode

The offset persists but drifts, which we model as a random walk over the arrivals of admissible messages from one sender, $\boldsymbol{\delta}_t = \boldsymbol{\delta}_{t-1} + \mathbf{w}_t$, $\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, q^2\mathbf{I})$. Together with Eq. (4) this is a linear-Gaussian model, so the posterior given all past anchors is Gaussian.

Proposition 1 (Scalar recursion)**.** *If the prior covariance, the process noise and the measurement noise are all isotropic, the posterior covariance stays isotropic,* $\mathbf{P}_t = p_t\mathbf{I}$*, and the Kalman filter reduces exactly to*

$$p_t^- = p_{t-1} + q^2, \qquad k_t = \begin{cases} \dfrac{p_t^-}{p_t^- + r^2/n_t}, & n_t \geq 1, \\ 0, & n_t = 0, \end{cases} \tag{5}$$

$$\hat{\boldsymbol{\delta}}_t = \hat{\boldsymbol{\delta}}_{t-1} + k_t(\bar{\mathbf{z}}_t - \hat{\boldsymbol{\delta}}_{t-1}), \qquad p_t = (1 - k_t)\, p_t^-. \tag{6}$$

The proof is a two-line induction (Section A). Each stream starts from $\hat{\boldsymbol{\delta}}_0 = \mathbf{0}$ and $p_0 = 1\,\mathrm{m}^2$. A message with many anchors receives a large gain; a message without anchors leaves the estimate unchanged while p_t grows; the first anchors after a gap are weighted strongly, so the estimate catches up quickly after an occlusion (Fig. 3).

Between registration and calibration. Equation (6) is an exponentially weighted average $\hat{\boldsymbol{\delta}}_t = (1-k_t)\hat{\boldsymbol{\delta}}_{t-1} + k_t\bar{\mathbf{z}}_t$ whose gain is set by q/r, the anchor count and the time since the last anchor. Its two limiting cases are the classical answers to pose error: $q \to \infty$ gives $k_t \to 1$ and $\hat{\boldsymbol{\delta}}_t = \bar{\mathbf{z}}_t$, i.e., per-frame registration from the current correspondences; $q \to 0$ gives $k_t \to 0$ after convergence, i.e., a fixed calibration averaged over the stream. For a stream with n anchors per message the gain converges to

$$k_\infty = \frac{p_\infty^-}{p_\infty^- + r^2/n}, \qquad p_\infty^- = \tfrac{1}{2}\Big(q^2 + \sqrt{q^4 + 4q^2r^2/n}\Big). \tag{7}$$

With $q = 0.05\,\mathrm{m}$, $r = 0.6\,\mathrm{m}$ and the 4.79 anchors that an anchor-bearing validation message holds on average, $k_\infty \approx 0.17$: the estimate pools roughly the last six anchor-bearing messages, about one second at 10 Hz once the anchor-free messages are included. That is long enough to average out object-level noise and short enough to follow the drift.

Hyper-parameters. We fix $r = 0.6\,\mathrm{m}$ and $g = 2.5\,\mathrm{m}$ for all detectors and select $q \in \{0.025, 0.05, 0.1, 0.15\}\,\mathrm{m}$ per detector on the calibration split only ($0.05\,\mathrm{m}$ for the released TraF-Align boxes; Section C). Streams are kept per sender; in our offline replay each latency budget is an independent replay, so the stream key is (sequence, budget).

Transfer. The tracked offset corrects *every* remote box, $\hat{\mathbf{x}}_j = \tilde{\mathbf{x}}_j + \hat{\boldsymbol{\delta}}_t$, including the roadside-only objects that have no ego counterpart. Because $\hat{\boldsymbol{\delta}}_t$ persists, messages without any anchor are corrected as well. This is the essential difference from per-frame registration, which returns no correction whenever no correspondence is found and therefore gives up exactly on the frames in which the partner's evidence matters most.

Algorithm 1: CoMET on the receiver at decision time t

Input: ego boxes $\mathcal{B}_e(t)$; admissible messages $m_\tau, m_{\tau'}$; online pose $\mathbf{T}_{e\leftarrow i}(t)$; stream state $(\hat{\boldsymbol{\delta}}, p)$
Output: ranked detections $\mathcal{D}_t$

$\{\tilde{\mathbf{b}}_j\} \leftarrow$ propagate and map m_τ by Eq. (2);
if m_τ *is new* **then** // once per message
  $p \leftarrow p + q^2$; $\mathcal{A} \leftarrow$ anchors of all ego boxes by Eq. (3);
  if $n = |\mathcal{A}| \geq 1$ **then**
    $\bar{\mathbf{z}} \leftarrow \frac{1}{n}\sum_{(k,j)\in\mathcal{A}}(\mathbf{x}_k^{\text{ego}} - \tilde{\mathbf{x}}_j)$; $k \leftarrow p/(p + r^2/n)$;
    $\hat{\boldsymbol{\delta}} \leftarrow \hat{\boldsymbol{\delta}} + k(\bar{\mathbf{z}} - \hat{\boldsymbol{\delta}})$; $p \leftarrow (1-k)\,p$;
$\hat{\mathbf{b}}_j \leftarrow \tilde{\mathbf{b}}_j$ with centre $\tilde{\mathbf{x}}_j + \hat{\boldsymbol{\delta}}$ for all j; // transfer
$\mathcal{E}^+ \leftarrow \{\mathbf{b}_k \in \mathcal{B}_e(t) : s_k \geq \tau_p\}$; $\mathcal{C} \leftarrow \{\hat{\mathbf{b}}_j\} \cup (\mathcal{B}_e(t) \setminus \mathcal{E}^+)$;
$u_c \leftarrow f_\phi(\mathbf{f}_c)$ for $c \in \mathcal{C}$; // frozen readout head
$\mathcal{D}_t \leftarrow \mathcal{E}^+$ ranked by $1 + s_k$, followed by $\mathcal{C}$ ranked by u_c;

4.4 Re-acquiring a lost lock

The gate g makes the association robust, but it also bounds the offsets that CoMET can lock onto. If $\|\boldsymbol{\delta}_t\|$ exceeds g when a stream starts, or jumps beyond it later, no co-visible pair falls inside the gate and the filter never receives an observation. A lock-acquisition step removes this limitation; its parameters are fixed in advance and are not tuned.

A stream is declared *lost* after $K = 5$ consecutive messages in which both agents report at least three boxes but no anchor is found. The receiver then collects all ego–remote displacements $\mathbf{x}_k^{\text{ego}} - \tilde{\mathbf{x}}_j$ of norm at most 10 m from the last $W = 5$ messages and votes them into a two-dimensional histogram with 0.5 m bins. The coarse offset $\mathbf{c}$ is the mean of the displacements within 1 m of the highest bin. It is accepted only if at least six pairs from at least three messages support it, if $\|\mathbf{c}\| \geq 2$ m (smaller offsets are within reach of the gate), and if the peak holds at least 1.5 times the count of the highest non-adjacent bin; otherwise the vote is repeated with the next message. On acceptance the filter is reset to $\hat{\boldsymbol{\delta}} = \mathbf{c}$ and $p = 0.25\,\text{m}^2$, and association continues on the remote centres shifted by the fixed $\mathbf{c}$, with observation $\mathbf{c} + \bar{\mathbf{z}}_t$ in Eq. (6). Because $\mathbf{c}$ is held fixed instead of following the running estimate, the filter cannot hide the anchors that would correct it, which is the failure mode of prediction gating (Section 6.4). A later loss of lock triggers a new acquisition.

As long as no stream is lost, the step is inactive and CoMET is unchanged. On the real data of this paper it changes validation AP@0.7 by less than 0.01 points for every detector (Section 6.10); unless stated otherwise, the reported results are those without it.

4.5 Late fusion and readout

Correct geometry is necessary but not sufficient: a corrected box only helps if it outranks the false positives it competes with, and the scores of two detectors are not comparable. We therefore rank the confident ego detections $\mathcal{E}_t^+ = \{k : s_k \geq \tau_p\}$ first and score every other candidate, i.e., corrected remote boxes and low-score ego boxes, with a small frozen readout head. The head is a two-layer perceptron with 32 hidden units (1.3K parameters) on 39 receiver-side features such as message age, box geometry and the overlap with the nearest ego detection; the 13 features that encode absolute position, including the sensor-to-sensor translation, are set to zero so that the head cannot learn location priors (Section C). It is trained once with binary cross-entropy on the fit split to predict whether a candidate adds a correct detection (IoU ≥ 0.7 with a ground-truth box not yet matched by $\mathcal{E}_t^+$), and then kept fixed; it never sees the validation or held-out splits. τ_p is selected per detector

on the calibration split, and whether to keep the original heads or refit them on CoMET geometry is decided once, globally, on the calibration split (the original heads are kept). The pose estimator itself has no learned parameters.

Cost. A message carries boxes, scores and the capture time. Per message the receiver solves one assignment between a few dozen boxes, applies the closed-form update of Eq. (6) and evaluates the 1.3K-parameter head on each candidate; the state is two coordinates and one variance per sender.

4.6 Second tier: retraining intermediate fusion

CoMET produces a causal relative pose, so it can also replace the online pose of an intermediate-fusion model. We refer to the pipeline above, with its training-free pose estimate and frozen readout, as the first tier. When an intermediate-fusion model is retrained anyway, we change two things and nothing else. *Pose input:* the fusion network receives the remote features warped with the CoMET estimate at training and at test time. *Training labels:* on placeholder messages of the fit split, the roadside-only ground truth is re-registered with the per-sequence linear interpolation of the non-zero dataset offsets, so that the network is not trained to reproduce the uncorrected projection; this uses labels and is applied to training only. We also tested a variant that additionally moves roadside-only labels on saturated axes to the median co-labelled displacement; the calibration split preferred the simpler variant (Section E). Evaluation always uses the official ground truth.

Combining the two tiers. The two tiers err differently: the first places boxes more precisely, and the retrained network finds more objects (Section 6.8). A box-level combination keeps the strengths of both. Boxes of the two tiers are paired one-to-one by BEV IoU, greedily in decreasing IoU, if the IoU is at least 0.3. A pair keeps the first-tier box and the larger of the two scores. An unpaired box keeps its geometry, and its score is multiplied by a per-tier factor α. Overlapping boxes are finally removed by BEV non-maximum suppression at IoU 0.15. The factor α is the least-squares scale that maps a tier's scores on the calibration split to their correctness at IoU 0.7, clipped to $[0, 1]$; it is 0.39 for the first tier and 1 for the second, so first-tier boxes that the network does not confirm are down-weighted. The pairing threshold (0.3 or 0.5) and whether the scores are first calibrated by isotonic regression were chosen among four settings on the calibration split. The combination uses no ground truth at inference.

5 Experimental Protocol

Data and splits. We use the sequential V2I perception subset of V2X-Seq [29]. Cooperative labels of the test split are not public, so all reported numbers are on the 21 official validation sequences (3,316 messages at each budget, 31,123 vehicle boxes), which no detector has seen during training and which are never used for any selection. The 46 training sequences are divided into a *fit* split (24) for readout heads and retraining, a *calibration* split (8) that is the only split used for selecting any hyper-parameter, and a *held-out* split (14) that we report as a secondary result. The held-out split lies inside the detectors' training data; results on it are reported but never used to choose anything.

Protocol and metric. We use the strict arrival protocol of Section 3.1 with $d \in \{0, 100, 200, 400\}$ ms. We report bird's-eye-view (BEV) average precision at IoU 0.7 and 0.5 (AP@0.7, AP@0.5) for vehicles in the common region $x \in [0, 96]$ m, $|y| \leq 38.4$ m of the ego

frame, with predictions ranked globally over the split. Unless stated otherwise a number is the mean over the four budgets and over the three frozen readout heads (seeds 2023/2024/2025) or three training seeds.

Statistics. Confidence intervals are 95% whole-sequence paired bootstrap intervals with 2,000 replicates [4]: sequences are resampled with replacement, both methods are evaluated on the same draw, and draws are paired before averaging over heads, seeds and budgets. We recomputed every reported number from the saved detections with a second, independently written implementation of box matching, AP and bootstrap; the two implementations agree to numerical precision. Where a table lists per-head intervals we give the envelope of the three heads' intervals.

Populations and auxiliary references. Following Section 3.4 we additionally split each result into *all* messages (primary), messages with an exactly *zero* offset, and messages with at least one *co-labelled* object. The populations are defined by annotations, used for reading only, and never as an input or for selection. The two auxiliary references move only roadside-only ground-truth boxes, keep all predictions, rankings and the evaluation region unchanged, and are used as sensitivity analyses: *interpolated* (placeholder messages re-registered with the per-sequence interpolation of non-zero offsets) and *label-median* (messages with at least three co-labelled objects re-registered with their median displacement; otherwise the official or interpolated offset).

Detectors and baselines. *Detectors.* For late fusion we use the single-agent boxes of five detectors from three architectures: the released TraF-Align [15] weights, TraF-Align retrained under the strict protocol, two V2X-ViT [20] models trained with different seeds, and Where2comm [5]. Unless stated otherwise, results use the released TraF-Align boxes, the strongest single-agent detector (62.15 AP@0.7). *Intermediate fusion.* F-Cooper [2] and V2X-ViT (three seeds) trained under the strict protocol, and TraF-Align with its released weights, all with online poses. *Late fusion.* The official DAIR-V2X late-fusion baseline [27] without and with its time compensation (we report the better of the two per column), and our late fusion with online poses and the same readout as CoMET. *Per-frame registration.* The official box-level modules of FreeAlign [10], RoCo [6] and CBM [14] on the same candidate pool, each followed by the same readout as CoMET; we additionally smooth each module's per-frame estimate with a temporal median or an innovation-gated Kalman filter whose parameters are selected on the calibration split. *Pose graph.* CoAlign [13] trained on V2X-Seq under our protocol with three seeds. In Section 3.2 we also show, as an upper reference that is not deployable, late fusion placed with the dataset offset, which uses annotations of the current frame.

Controlled pose errors. To study larger and changing errors on real data, we add a known planar translation to every remote box of a message before any processing, in one random direction per sequence that is shared by all methods and latency budgets; the ground truth is unchanged and no method has access to the injected value. We use constant translations of 0.5, 1, 2, 3 and 5 m, a drift that grows linearly from 0 to 2 m over the sequence, and a 2 m step at the temporal midpoint of the sequence. For intermediate fusion the same translation is added to the relative pose given to the network.

Specification in advance. The lock-acquisition step of Section 4.4 and the two-tier combination of Section 4.6 were specified, with their parameters and the criteria for adopting them, before the validation and held-out splits were evaluated. The acceptance criteria are given in Section F.

Table 2: V2X-Seq validation under strict arrival with online poses. AP@0.7 and AP@0.5 (%) per latency budget d and their mean. †Box-level methods on the released TraF-Align boxes with the same readout as CoMET (three frozen heads). ‡Temporal median of CBM's per-frame estimates; the best of six smoothed external variants on the calibration split.

	AP@0.7					AP@0.5				
Method	0	100	200	400	mean	0	100	200	400	mean
Ego detector only	62.15	62.15	62.15	62.15	62.15	70.93	70.93	70.93	70.93	70.93
F-Cooper [2]	62.31	62.20	62.20	62.01	62.18	77.33	77.10	76.85	76.48	76.94
TraF-Align [15]	60.04	61.00	61.16	62.26	61.11	75.11	75.37	75.18	75.41	75.27
V2X-ViT [20] (3 seeds)	63.11	62.93	62.91	62.91	62.96	78.31	77.98	77.64	77.32	77.81
Official late fusion [27] (best of two)	63.06	62.92	62.85	62.55	62.84	74.23	74.03	73.84	73.07	73.79
Late fusion, online pose†	63.21	63.18	63.14	63.18	63.18	74.45	74.33	74.25	74.06	74.27
FreeAlign [10]†	64.09	63.39	63.29	63.18	63.49	75.16	73.75	73.73	73.38	74.01
RoCo [6]†	65.49	64.33	63.78	63.84	64.36	77.01	75.25	74.09	74.02	75.09
CBM [14]†	65.26	64.10	63.67	63.21	64.06	76.43	74.82	73.72	72.85	74.46
CBM + temporal median†‡	65.26	64.01	63.60	63.14	64.00	76.58	74.82	73.66	72.94	74.50
CoMET (ours)†	**67.37**	**66.88**	**66.54**	**65.72**	**66.63**	**79.55**	**79.18**	**78.84**	**77.94**	**78.87**

Reproducibility. The estimator is re-implemented in a stand-alone package that reproduces the per-message offsets, variances and anchor counts behind all tables exactly (maximum absolute difference 0 over 205,202 messages, five detectors and four splits). The same package reproduces the lock-acquisition states of all real and injected-error runs and the combined detections of all 69,024 evaluated frames of the two-tier combination exactly. The code, the protocol and all result tables are released.

6 Results

6.1 Main comparison

Table 2 compares CoMET with intermediate fusion, late fusion and per-frame registration. Under online poses, intermediate fusion gains little: the three models range from 61.11 to 62.96 AP@0.7 against 62.15 for the ego detector alone, although they do detect more objects at the loose threshold (up to 77.81 AP@0.5 against 70.93). They find the hidden objects but cannot place them precisely enough. Late fusion with online poses behaves similarly at AP@0.7 (63.18). Per-frame registration helps, confirming that geometry is the bottleneck, but each module re-estimates the pose in every frame from the current correspondences: the best (RoCo) reaches 64.36 AP@0.7, and smoothing the per-frame estimates over time does not close the gap (Section 6.5).

CoMET reaches 67.37/66.88/66.54/65.72 AP@0.7 at 0/100/200/400 ms, 66.63 on average: 3.45 points above late fusion with the same readout and online poses, 3.67 above the best intermediate fusion and 2.27 above the best per-frame registration module. It also has the highest AP@0.5 of the deployable methods in Table 2 (78.87 on average), so the gain is not bought by trading recall for precision. Relative to placing the boxes with the dataset's annotation-assisted offset (69.52, Section 3.2), which is not available online, CoMET closes 54% of the gap left by online poses without using any annotation.

Table 3: CoMET on five detectors. AP@0.7 (%) of CoMET averaged over budgets and readout heads, and its paired difference to late fusion with the online pose (column: online) and to estimating the offset from the current message only (column: single), with the same readout. Every per-head 95% interval of the difference to the online pose lies above zero; see Table 9 for intervals.

	validation			held-out		
Detector	CoMET	−online	−single	CoMET	−online	−single
TraF-Align (released)	66.63	+3.45	+0.57	72.57	+8.37	+1.38
TraF-Align (strict retrain)	58.96	+2.74	+0.30	61.57	+6.82	+1.16
V2X-ViT, seed A	66.08	+2.38	+0.49	71.30	+5.92	+1.24
V2X-ViT, seed B	64.97	+2.22	+0.42	68.95	+4.12	+0.83
Where2comm	65.14	+3.05	+0.46	71.29	+6.73	+1.19

Table 4: How the per-message anchor evidence is aggregated over time. Released TraF-Align boxes, same anchors and readout, mean over budgets and heads.

	validation		held-out	
Estimator of $\boldsymbol{\delta}_t$	AP@0.7	AP@0.5	AP@0.7	AP@0.5
Current message, mean	66.06	78.17	71.19	85.93
Current message, median	66.05	78.14	71.29	85.90
Hold last observation	65.99	78.26	71.05	85.80
Exponential moving average	66.17	78.50	72.49	87.03
Sliding-window mean	66.63	78.71	72.07	86.77
Common-mode KF (CoMET)	66.63	78.87	72.57	86.99

6.2 Across detectors

Table 3 applies CoMET to the boxes of all five detectors, with q and τ_p selected per detector on the calibration split and every other setting shared. On the validation split CoMET improves on late fusion with online poses by 2.2–3.5 AP@0.7 and on per-message estimation by 0.3–0.6 for every detector; on the held-out split the improvements are larger (4.1–8.4 and 0.8–1.4). The improvement is not specific to one detector or architecture, and with V2X-ViT or Where2comm boxes CoMET reaches 65.0–66.1 AP@0.7, so the result in Table 2 does not stem from a stronger single-agent detector.

6.3 Why tracking: temporal aggregation

Table 4 keeps the anchors fixed and changes only how they are aggregated. Using the current message alone, by its mean or median, gives 66.06 AP@0.7; holding the last observation through anchor-free messages does not help. Pooling over time helps on both splits, and the Kalman filter is the best or tied for best everywhere: it matches the sliding window on the validation split (66.63) and is ahead of it on the held-out split (72.57 against 72.07), where the exponential average is close (72.49). The filter is preferable to fixed windows for a second reason: its gain adapts to the number of anchors and to the time since the last one, which a window or a constant-gain average cannot do. On the held-out split the difference between the filter and per-message estimation reaches 1.38 AP@0.7.

Table 5: Association and anchor choices. AP@0.7 (%) averaged over the five detectors, three heads and four budgets. All variants share one fixed setting of q and τ_p per detector, which precedes the final per-detector selection, so absolute values differ from Table 2; the calibration column is the selection criterion. Confident ego anchors: ego detections with score $\geq \tau_p$. Prediction-gated: remote boxes are shifted by the current estimate before matching.

Anchor candidates	Association	calibration	validation	held-out
confident ego anchors (score $\geq \tau_p$)	Hungarian + mutual NN	66.39	64.15	68.94
confident ego anchors	Sinkhorn	65.72	64.12	68.59
confident ego anchors	Hungarian, prediction-gated	66.02	64.14	68.34
all ego boxes	Sinkhorn, σ=1 m, centred on estimate	66.12	64.40	68.79
all ego boxes	Sinkhorn, σ=1 m, uncentred	66.28	63.92	68.04
all ego boxes	Sinkhorn, σ=0.5 m	65.87	64.16	69.09
all ego boxes	Sinkhorn, σ=2 m	65.92	63.91	67.45
all ego boxes	learned soft assignment	66.24	64.41	68.69
all ego boxes	Hungarian, prediction-gated (2.5 m)	66.24	64.41	68.40
all ego boxes	Hungarian, prediction-gated (1.5 m)	66.30	64.44	68.56
all ego boxes (CoMET)	**Hungarian + mutual NN**	66.89	64.36	69.06

Table 6: Sensitivity (report only, never used for selection). Released TraF-Align boxes, validation AP@0.7. Anchors are restricted to ego detections above the score threshold; "all" uses every ego detection.

Anchor score threshold	all	0.1	0.2	0.3	0.4	0.5
AP@0.7	66.63	66.63	66.63	66.50	66.25	65.99
Association gate (m)	1.5	2.0	2.5	3.0		
AP@0.7	66.21	66.65	66.63	66.66		

6.4 Anchors, association and learned alternatives

Table 5 varies the anchor set and the association while keeping the filter. Two findings matter. *Use all ego detections.* Restricting anchors to confident ego detections (first three rows) loses evidence; with all ego boxes and hard mutual-nearest-neighbour matching (last row, the configuration used throughout) the calibration score is the highest of all variants. Consistently, raising an anchor score threshold only lowers AP@0.7 (Table 6), and the result is flat for gates between 2.0 and 3.0 m. *Structure beats capacity.* Soft assignment with Sinkhorn normalisation [3] at various widths and a learned soft assignment trained end to end do not improve on hard matching: on the validation split all variants lie within 0.6 points of each other, on the calibration split hard assignment is the best, and on the held-out split it is within 0.03 points of the best. Gating the association with the filter prediction is worse on the calibration and held-out splits: when the current offset is large, the prediction hides exactly the anchors that would reveal it. We therefore keep the simplest configuration that wins on the calibration split.

6.5 Per-frame registration with temporal smoothing

One could argue that the advantage of CoMET comes from temporal smoothing alone, which could be added to any registration module. Table 7 tests this by smoothing the per-frame estimates of FreeAlign, RoCo and CBM with a temporal median or an innovation-gated Kalman filter, with parameters selected on the calibration split. Smoothing changes their AP@0.7 by at most 0.4 points; the variant selected on the calibration split (CBM with a three-message median) reaches 64.00, 2.62

Table 7: External registration modules with temporal processing. Validation AP@0.7 (%), released TraF-Align boxes, same readout as CoMET. Median: temporal median over the last messages; gated KF: Kalman filter on the module's per-frame estimate with innovation gating. Window length and gate are selected on the calibration split. CoMET: 66.63.

Module	per frame	+ temporal median	+ gated KF
FreeAlign	63.49	63.54	63.88
RoCo	64.36	64.30	64.20
CBM	64.06	64.00	63.88

Table 8: CoAlign with different relative poses. Full CoAlign fusion, three training seeds, common evaluation region. AP@0.7 per budget and mean, and mean AP@0.5.

Relative pose	0	100	200	400	mean	AP@0.5
CoAlign pose graph [13]	56.99	56.64	56.47	56.26	56.59	72.30
CoMET, confident anchors only	57.95	57.74	57.52	57.21	57.60	73.67
CoMET	58.06	57.82	57.59	57.33	57.70	73.74

points below CoMET. A plausible reason is that per-frame registration from object graphs or local context fails grossly when few objects are shared, and smoothing a sequence that contains gross failures does not produce a good estimate, whereas the mutual-nearest-neighbour anchor observation of CoMET rarely fails grossly (Section 3.3).

6.6 CoMET as the pose of a pose-graph system

CoAlign [13] optimises an agent–object pose graph in every frame before multi-scale feature fusion. Feeding CoMET's causal pose to the CoAlign fusion network instead of the pose-graph output improves AP@0.7 by 1.11 points (57.70 against 56.59, consistently for all three seeds) and AP@0.5 by 1.44 (Table 8). Restricting the anchors of CoMET to confident ego detections lowers this gain by only 0.10 points, so for intermediate fusion the main benefit comes from tracking itself rather than from the choice of anchors. CoAlign's per-frame graph cannot use evidence from earlier messages; CoMET can.

6.7 Robustness to the reference

Populations. Table 9 splits the official result. On messages with co-labelled objects, where the official ground truth of every object is registered, CoMET improves more than on all messages, by 2.9–4.5 points over online poses and 0.5–1.0 over per-message estimation, and all per-head intervals of both differences lie above zero for every detector. On placeholder messages the official ground truth of roadside-only objects is the uncorrected projection, so moving these boxes to where the objects are is counted as an error there; the auxiliary references below show that this effect is a label artefact. We therefore do not use any rule that withholds the correction on frames without anchors, which would gain on the official table only through the placeholder.

Auxiliary references. Table 10 repeats the main comparisons under the two auxiliary references. With either reference the improvement over online poses grows (validation: +4.0 and +3.9 for the released TraF-Align boxes) and the improvement over per-message estimation stays positive with intervals above zero on both splits and both detectors (+0.35 to +1.58). On placeholder

Table 9: Results by label-defined population, validation AP@0.7, paired differences in points (mean over three heads). "All" is the official result; "co-lab." are messages with a co-labelled object, on which the official ground truth of every object is registered; "zero" are placeholder messages, on which the official ground truth of roadside-only objects is the uncorrected projection.

	CoMET − online pose			CoMET − single message		
Detector	all	co-lab.	zero	all	co-lab.	zero
TraF-Align (released)	+3.45	+4.53	−3.42	+0.57	+0.98	−2.02
TraF-Align (strict)	+2.74	+3.47	−2.01	+0.30	+0.53	−1.10
V2X-ViT A	+2.38	+3.16	−2.61	+0.49	+0.62	−0.28
V2X-ViT B	+2.22	+2.93	−2.32	+0.42	+0.55	−0.35
Where2comm	+3.05	+3.97	−2.73	+0.46	+0.57	−0.15

Table 10: Auxiliary references (sensitivity; not physical truth). AP@0.7 differences with 95% intervals, mean over three heads. Interpolated: placeholder messages re-registered with the per-sequence interpolation of non-zero offsets (released TraF-Align boxes). Label-median: messages with at least three co-labelled objects re-registered with their median displacement.

		interpolated	label-median	
Split	Difference	TraF-Align (released)	TraF-Align (released)	TraF-Align (strict)
validation	CoMET − online pose	+4.00 [2.41, 5.69]	+3.93 [2.26, 5.78]	+2.84 [1.66, 4.08]
validation	CoMET − current message	+0.94 [0.58, 1.33]	+0.90 [0.50, 1.35]	+0.35 [0.09, 0.60]
held-out	CoMET − online pose	+8.73 [2.51, 12.52]	+9.30 [2.65, 13.43]	+7.32 [1.95, 10.57]
held-out	CoMET − current message	+1.58 [0.18, 2.58]	+1.42 [0.05, 2.42]	+1.10 [0.16, 1.78]

messages the sign reverses: once the roadside-only objects there are registered, CoMET is better than the online pose (released +0.83, strict +0.38 on validation, both intervals above zero; Table 18), confirming that the official penalty on these messages is a label artefact. The conclusions of Tables 2 and 4 therefore do not depend on the label artefacts.

6.8 Second tier: retraining intermediate fusion

Table 11 retrains the fusion network and detection head of TraF-Align for a fixed budget, changing only the pose and, optionally, the training labels. Compared with retraining under the online pose, the second tier improves validation AP@0.7 by 0.81 and AP@0.5 by 1.17 points, both with intervals above zero. Its AP@0.5 of 81.44 is the highest of all methods on the validation split, 2.56 points above CoMET without retraining. At AP@0.7 the retrained network is on par with CoMET without retraining (−0.14, interval including zero), so retraining mainly adds recall. Retraining under a corrected pose accounts for most of the improvement over the online pose, and cleaned training labels add a further 0.25 AP@0.7 (66.18 to 66.44 with a CoMET pose whose anchors are restricted to confident ego detections); held-out results and the training budget are in Section E.

Table 12 applies the same recipe to Where2comm and CoAlign. The second tier improves on retraining with online poses for both: clearly for CoAlign (+1.63 AP@0.7, +1.75 AP@0.5, intervals above zero) and slightly for Where2comm (+0.24, +0.32). The second tier is thus a drop-in improvement for systems that retrain their fusion network anyway, while the first tier, which needs

Table 11: TraF-Align retrained from the released weights (fusion and head fine-tuned for 2,000 updates, reader and backbone frozen, three seeds), validation, mean over budgets and seeds. Top: absolute AP. Bottom: the second tier minus each alternative, with 95% intervals.

Training and test pose / training labels	AP@0.7	AP@0.5
Online pose, original labels	65.68	80.26
CoMET pose (confident anchors), original labels	66.18	81.19
CoMET pose (confident anchors), cleaned labels	66.44	81.36
CoMET pose, cleaned labels (second tier)	66.49	81.44
CoMET without retraining (first tier)	66.63	78.87

Second tier minus	AP@0.7	AP@0.5
retraining with the online pose	+0.81 [0.18, 1.62]	+1.17 [0.58, 1.87]
CoMET without retraining (first tier)	−0.14 [−1.29, 0.88]	+2.56 [1.12, 3.98]

Table 12: The same recipe on two more architectures, validation, three seeds. AP@0.7/ AP@0.5 of the fusion network retrained with the online pose, with a CoMET pose whose anchors are restricted to confident ego detections, and with CoMET poses and cleaned labels (second tier); last two columns: second tier minus the online pose with 95% intervals.

Architecture	online	confident anchors	second tier	ΔAP@0.7	ΔAP@0.5
Where2comm	63.05 / 76.96	63.03 / 77.07	63.29 / 77.28	+0.24 [−0.16, 0.54]	+0.32 [−0.29, 0.84]
CoAlign	56.47 / 72.13	57.85 / 73.78	58.10 / 73.88	+1.63 [1.20, 2.10]	+1.75 [1.03, 2.62]

no retraining, remains the default.

6.9 Combining the two tiers

The first tier is better at AP@0.7 and the second at AP@0.5. Table 13 shows that the box-level combination of Section 4.6 is better than both on both metrics. On the validation split it reaches 67.85 AP@0.7 and 82.62 AP@0.5, the highest values of all methods in this paper. This is 1.23 AP@0.7 above the first tier and 1.36 above the second, and 1.18 AP@0.5 above the second tier, all with intervals above zero. On the held-out split it improves AP@0.7 over both tiers as well (73.28; +0.71 and +1.84). The combination needs no training beyond the two tiers. Its only fitted quantities are the two score factors: 1 for unpaired second-tier boxes and 0.39 for unpaired first-tier boxes, which down-weights first-tier boxes that the retrained network does not confirm.

6.10 Large and changing pose errors

The real offsets of V2X-Seq rarely exceed the association gate, so we inject known errors to test larger and changing ones (Section 5). Figure 5a shows that CoMET follows constant errors up to 1 m, losing at most 0.44 AP@0.7. At 2 m it loses 1.4–1.7 points as anchors start to fall outside the gate, and beyond the 2.5 m gate it can no longer lock and falls nearly to the level of the online pose. With lock acquisition, the accuracy stays almost flat: at 3 and 5 m the step adds 1.4–4.5 AP@0.7 over CoMET for all five detectors, with intervals above zero, and retains 97.6–99.5% of CoMET's accuracy without injected error (Table 14). Without injected error the step changes validation AP@0.7 by less than 0.01 points for every detector; it is triggered nine times on the 420 validation streams of the five detectors and never on the held-out split. Changing errors are already

Table 13: Box-level combination of the two tiers (Section 4.6). TraF-Align boxes; mean over four budgets and three heads (first tier) or three training seeds (second tier), paired by position. Bottom: paired differences with 95% intervals.

	validation		held-out	
	AP@0.7	AP@0.5	AP@0.7	AP@0.5
CoMET late fusion (first tier)	66.63	78.87	72.57	86.99
Retrained TraF-Align (second tier)	66.49	81.44	71.45	89.59
Two-tier combination	**67.85**	**82.62**	**73.28**	**90.01**
Combination − first tier	+1.23 [0.42, 2.06]	+3.75 [2.20, 5.34]	+0.71 [0.49, 1.04]	+3.02 [2.24, 4.00]
Combination − second tier	+1.36 [0.71, 2.18]	+1.18 [0.67, 1.70]	+1.84 [0.25, 2.53]	+0.42 [−0.06, 1.63]

Table 14: Injected constant pose error (Section 5), validation AP@0.7 (%). Late fusion with the online pose, CoMET, and CoMET with lock acquisition (Section 4.4); paired difference of the last two with 95% interval, and the share of CoMET's AP@0.7 without injected error that is retained.

Detector	Error	Online	CoMET	+ acquisition	Δ	Retained
TraF-Align (released)	3 m	61.86	62.70	**65.63**	+2.93 [1.31, 4.83]	98.5%
	5 m	61.77	61.84	**66.29**	+4.45 [2.85, 6.16]	99.5%
TraF-Align (strict)	3 m	55.19	55.89	**57.62**	+1.73 [0.71, 2.96]	97.7%
	5 m	55.07	55.15	**58.62**	+3.48 [2.30, 4.73]	99.4%
V2X-ViT A	3 m	62.53	63.15	**64.59**	+1.44 [0.62, 2.47]	97.7%
V2X-ViT B	3 m	61.72	62.28	**63.72**	+1.44 [0.68, 2.41]	98.1%
Where2comm	3 m	60.98	61.60	**63.58**	+1.97 [0.92, 3.30]	97.6%

followed by tracking: with a drift to 2 m or a 2 m step, CoMET loses at most 0.76 AP@0.7 and lock acquisition adds 0.01–0.13.

The corrected pose also protects intermediate fusion. Given the injected pose, the second-tier network reaches only 61.91–62.11 AP@0.7 at 1–3 m. Giving it instead the pose corrected by CoMET with lock acquisition, estimated from the late-fusion boxes of the same injected messages, improves AP@0.7 by 3.90–4.41 points, and by 2.15–2.74 for the network retrained with online poses, with all intervals above zero (Fig. 5b).

6.11 Controlled study on OPV2V

In simulation we can choose the error. Table 15 perturbs the partner pose of OPV2V [21] in three ways. With a structured error whose statistics are fitted to V2X-Seq, CoMET recovers 8.9 of the 11.9 AP@0.7 points lost at $d = 0$ and 3.8 of 7.7 at 400 ms. With independent per-frame noise, which has no persistent component, the stale boxes are already close to the oracle and CoMET still adds 0.6 and 1.8 points. With an ego-pose timing error, which produces a persistent offset during motion, CoMET recovers 2.7–5.8 points at $d = 0$. CoMET helps in proportion to how persistent and how shared the error is.

7 Discussion

Reading V2X-Seq results with online pose correction. Our audit changes how such results should be read. The dataset offset is the right reference for the official ground truth, because the ground truth is built from it; it is not a measurement of the physical pose error, being zero where it

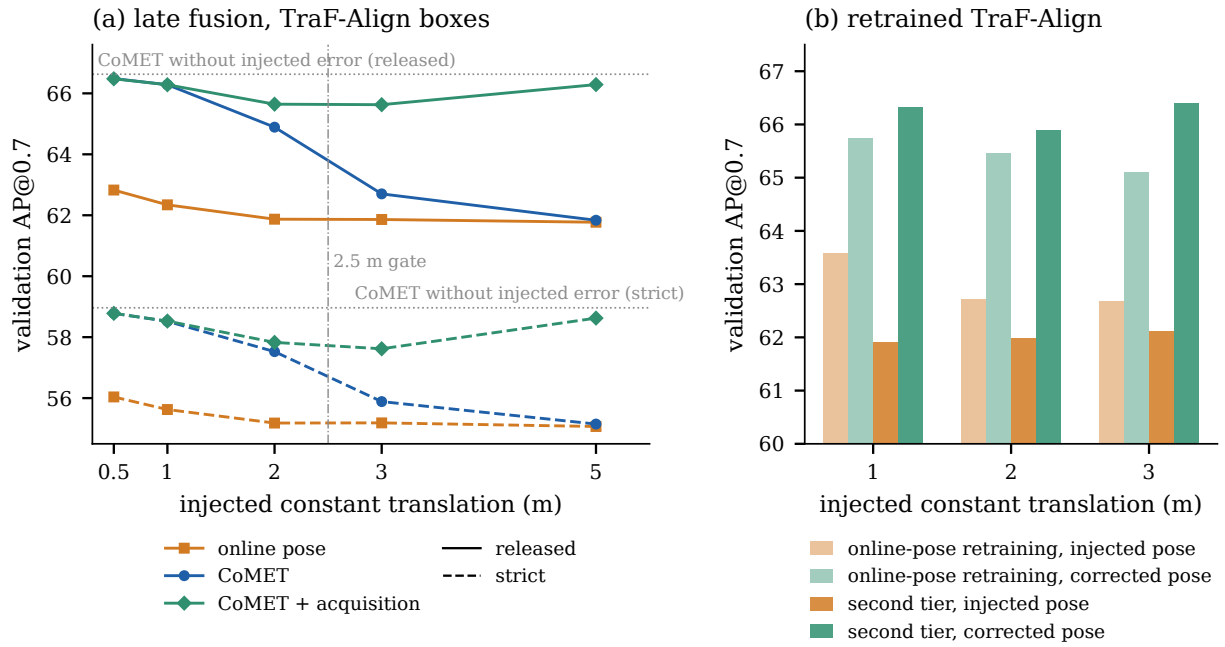


Figure 5: Validation AP@0.7 under an injected constant translation of the roadside message. (a) Late fusion with the released (solid) and strictly retrained (dashed) TraF-Align boxes. Without acquisition, CoMET falls towards the online pose once the error exceeds the 2.5 m association gate. (b) TraF-Align retrained with online poses and the second-tier network, given the injected pose or the pose corrected by CoMET with lock acquisition (one training seed).

could not be estimated and clipped at the bounds of its search. A method that estimates the true error is therefore counted as wrong on placeholder frames and on saturated frames. Population splits and auxiliary references do not replace physical ground truth, but they show which conclusions depend on these artefacts; for CoMET, none of these conclusions changes. We hope that future releases document how such offsets are produced and flag placeholders explicitly.

Limitations. CoMET assumes that one translation dominates the error of a message and that the objects seen by both agents share it with those seen only by the partner. When this does not hold, for example when the partner's objects are affected by errors that the co-visible objects do not reflect, the transferred correction can misplace boxes; a consistency check on the normalised filter innovation is a natural safeguard. The model is planar and translation-only, and the estimate follows abrupt changes of the offset with a short delay. Offsets larger than the association gate are recovered by lock acquisition only after a few messages with enough co-visible objects; until then the stream remains uncorrected. Our real-data evaluation uses one V2I dataset, complemented by a controlled simulation, and all numbers come from offline replay of recorded messages.

8 Conclusion

On real V2I data with online poses and strict arrival, the main obstacle to cooperative detection is a slowly drifting translation shared by all objects of a message, while the age of the message costs far less. Tracking this translation from co-visible anchors with a causal scalar filter, and transferring it to the objects only the partner sees, turns box messages into a clear gain over intermediate fusion

Table 15: OPV2V with controlled partner pose error. AP@0.7 (%), mean over three readout heads. Stale: remote boxes placed with the perturbed pose; oracle: the true pose.

	$d = 0$			$d = 400\,\text{ms}$		
Pose error	stale	CoMET	oracle	stale	CoMET	oracle
structured, fitted to V2X-Seq	73.22	82.14	85.07	66.90	70.73	74.62
i.i.d. per frame (0.2 m, 0.2°)	80.50	81.10	85.07	67.43	69.19	74.62
ego pose time error 50 ms	77.70	80.41	85.07	67.91	72.58	74.62
ego pose time error 100 ms	70.82	76.36	85.07	67.64	71.35	74.62
ego pose time error 150 ms	68.58	74.42	85.07	67.56	70.49	74.62

for every detector we tested, without training the estimator. With lock acquisition the tracker also recovers offsets of several metres, and the same pose improves intermediate fusion when it is retrained; combining both tiers at the box level gives the best accuracy at both IoU thresholds. We further find that the benchmark's own reference offset is a placeholder on a quarter of the frames and saturated on almost half of the rest, which matters for evaluating any geometric correction. The estimator, protocol and experimental results accompany the manuscript at https://github.com/sswun/CoMET-V2X. Use of the code or results requires prior written permission from the authors. The repository gives the terms and contact details.

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Appendix

A Proof of the scalar recursion

Let the state $\boldsymbol{\delta}_t \in \mathbb{R}^2$ follow $\boldsymbol{\delta}_t = \boldsymbol{\delta}_{t-1} + \mathbf{w}_t$ with $\mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, q^2\mathbf{I})$, and let a message with $n_t \geq 1$ anchors provide $\bar{\mathbf{z}}_t = \boldsymbol{\delta}_t + \bar{\boldsymbol{\eta}}_t$ with $\bar{\boldsymbol{\eta}}_t \sim \mathcal{N}(\mathbf{0}, \frac{r^2}{n_t}\mathbf{I})$; the mean of n_t independent residuals with covariance $r^2\mathbf{I}$ has covariance $\frac{r^2}{n_t}\mathbf{I}$, and it is a sufficient statistic for $\boldsymbol{\delta}_t$ because the likelihood of the individual residuals depends on them only through their mean. Assume $\mathbf{P}_{t-1} = p_{t-1}\mathbf{I}$, which holds for $t = 1$ with $p_0 = 1\,\mathrm{m}^2$. The prediction gives $\mathbf{P}_t^- = (p_{t-1} + q^2)\mathbf{I} = p_t^-\mathbf{I}$. With observation matrix $\mathbf{I}$ the Kalman gain is $\mathbf{K}_t = \mathbf{P}_t^-(\mathbf{P}_t^- + \frac{r^2}{n_t}\mathbf{I})^{-1} = \frac{p_t^-}{p_t^- + r^2/n_t}\mathbf{I} = k_t\mathbf{I}$, and the update gives $\mathbf{P}_t = (\mathbf{I} - \mathbf{K}_t)\mathbf{P}_t^- = (1 - k_t)p_t^-\mathbf{I}$. Without anchors there is no update and $\mathbf{P}_t = \mathbf{P}_t^-$. By induction the covariance stays isotropic and Eqs. (5) and (6) are exact. The steady state of Eq. (7) follows from the fixed point $p^- = (1-k)p^- + q^2$ with $k = p^-/(p^- + r^2/n)$, which gives $(p^-)^2 - q^2p^- - q^2r^2/n = 0$.

B Audit of the dataset offset

Procedure. For every training and validation frame pair we read the raw vehicle-side and roadside annotations, project the roadside labels into the vehicle frame with the public calibration, and join them to the cooperative labels by track and token. Cooperative labels carry a source flag (vehicle-only, roadside-only, co-labelled). For roadside-only objects we compare the cooperative label with the public projection, separately for frames with zero and non-zero offset; a second implementation repeats the join and the inverse transform chain independently. For co-labelled objects we compute the displacement between the vehicle label and the projected roadside label; its median over the objects of a frame (frames with at least three co-labelled objects) is the label-median displacement used in Section 3.4.

Findings in detail. The roadside-only cooperative label equals the public projection plus the offset, with a residual of about 4×10^{-7} m, for zero and non-zero offsets alike. Table 1 gives the census; in the fit, held-out and validation splits no zero message contains a co-labelled object, and only 10–16 non-zero messages lack one. The non-zero values per axis are $\{-1.10, -1.06, -1.02, -0.98, -0.94\}$, $\{-0.70, \ldots, -0.54\}$, $\{-0.30, \ldots, -0.14\}$, $\{+0.10, \ldots, +0.26\}$ and $\{+0.50, \ldots, +0.66\}$ in steps of 0.04 m, five values per group; together with zero these are 26 values. For interior values the median label-median displacement of the same frames differs from the value by at most about 0.1 m (Fig. 4); on the bounds the displacement continues beyond the bound. Over frames with at least three co-labelled objects and a non-zero offset, the correlation between offset and label-median displacement is 0.85 for x and 0.92 for y, and their median difference is 0.16 m; the offset is therefore close to, but not equal to, the label displacement, and we do not interpret it as a mean.

C Implementation details

Hyper-parameters. $r = 0.6$ m, $g = 2.5$ m, $p_0 = 1\,\mathrm{m}^2$ for all detectors. Selected on the calibration split: $q = 0.05/0.025/0.1/0.05/0.05$ m and $\tau_p = 0.5/0.3/0.5/0.5/0.5$ for the released TraF-Align, the strictly retrained TraF-Align, the two V2X-ViT models and Where2comm. The candidate grid for q was $\{0.025, 0.05, 0.1, 0.15\}$ m. The choice between the original readout heads and heads refitted on CoMET geometry was made once for all detectors on the calibration split (mean over five detectors 65.70 vs. 65.66 AP@0.7 for heads refitted after removing placeholder frames; original heads kept).

Anchor statistics. Table 16 lists the mean number of anchors per message and the fraction of messages without anchors for the released TraF-Align boxes.

Table 16: Mean anchors per message / fraction of messages without anchors, released TraF-Align boxes.

Split	0 ms	100 ms	200 ms	400 ms
fit	3.31 / 29.8%	3.28 / 30.0%	3.20 / 31.1%	2.98 / 33.1%
calibration	3.03 / 33.5%	2.97 / 34.6%	3.00 / 35.0%	2.89 / 34.2%
held-out	3.34 / 31.7%	3.35 / 30.8%	3.30 / 31.2%	3.11 / 32.3%
validation	2.96 / 38.2%	2.92 / 38.7%	2.90 / 37.6%	2.73 / 38.5%

Readout. The 39 receiver-side features of a candidate are: message age, detector score, box size, heading in the receiver and source frames, distance and BEV overlap to the nearest ego detection and that detection's score, the number of ego boxes within 10 m, flags for ego, propagated and causally associated candidates, propagation horizon, time between the two messages used for the velocity, speed, association distance (absolute and relative to the gate) and propagation shift. The 13 features that encode absolute position (receiver and source coordinates and ranges, and the source-to-receiver translation and range) are zeroed. The head is a two-layer perceptron (39–32–1, 1,313 parameters) trained with binary cross-entropy on candidates of the fit split; three heads with seeds 2023/2024/2025 are trained and all results average over them.

Second tier. Initial weights: the released TraF-Align weights (epoch 60). Reader and backbone are frozen (batch normalisation in evaluation mode); the fusion network, multi-scale aggregation and detection head are trained. AdamW, learning rate 10^{-4}, weight decay 0.01, gradient accumulation 4, 2,000 updates (8,000 micro-batches, about one pass over the 7,524 fit conditions at $d \in \{0, 400\}$ ms), no checkpoint selection on validation data. The loss is the original anchor classification and box regression loss on current cooperative ground truth. The same budget is used for the online-pose and CoMET arms and for Where2comm and CoAlign.

D Auxiliary references in detail

Table 17: Interpolated reference with three gap limits (released TraF-Align boxes, AP@0.7, 95% intervals). Placeholder runs are re-registered only if they are at most 1 s or 3 s long, or always.

	validation		held-out	
Re-registered	CoMET − online	CoMET − single	CoMET − online	CoMET − single
gaps ≤1 s	+3.66 [1.93, 5.48]	+0.60 [0.06, 1.13]	+8.43 [1.80, 12.41]	+1.43 [−0.15, 2.53]
gaps ≤3 s	+3.66 [1.93, 5.48]	+0.60 [0.06, 1.13]	+8.45 [1.89, 12.45]	+1.45 [−0.11, 2.55]
all gaps	+4.00 [2.41, 5.69]	+0.94 [0.58, 1.33]	+8.73 [2.51, 12.52]	+1.58 [0.18, 2.58]

E Second tier in detail

Held-out split. Table 19 gives the absolute AP of the retrained TraF-Align models on the held-out split (inside the detectors' training data). Tier 2 improves on retraining with online poses by +3.15

Table 18: Label-median reference by population (AP@0.7 differences, mean over three heads; every 95% interval excludes zero). On placeholder messages (zero) CoMET is now better than the online pose as well, because the reference registers the roadside-only objects.

		CoMET − online			CoMET − single		
Detector	Split	all	co-lab.	zero	all	co-lab.	zero
TraF-Align (released)	val.	+3.93	+4.45	+0.83	+0.90	+0.94	+0.77
TraF-Align (released)	held-out	+9.30	+10.21	+0.58	+1.42	+1.56	+0.22
TraF-Align (strict)	val.	+2.84	+3.24	+0.38	+0.35	+0.40	+0.13
TraF-Align (strict)	held-out	+7.32	+8.10	+0.58	+1.10	+1.22	+0.22

AP@0.7 [0.23, 4.81] and +1.91 AP@0.5.

Table 19: Retrained TraF-Align on the held-out split, mean over budgets and three seeds.

Training and test pose / training labels	AP@0.7	AP@0.5
Online pose, original labels	68.30	87.68
CoMET pose (confident anchors), original labels	72.15	90.09
CoMET pose (confident anchors), cleaned labels	72.73	89.94
CoMET pose, cleaned labels (second tier)	71.45	89.59
CoMET without retraining (first tier)	72.57	86.99

Label variants. The *interpolated* labels re-register roadside-only training labels on placeholder messages with the per-sequence interpolation of non-zero offsets. The *interpolated and bound-corrected* labels additionally move roadside-only labels on axes where the offset sits on a search bound to the label-median displacement of the frame (frames with at least three co-labelled objects). On the calibration split the first scored 67.91 and the second 66.66 AP@0.7 (three seeds, four budgets), so the interpolated labels were selected before the validation and held-out splits were evaluated; Table 20 reports both. The calibration split has the highest share of saturated messages (77.7%), on which the official ground truth favours predictions placed at the bound, which explains why the selection prefers the interpolated labels; we keep the variant selected in advance.

Table 20: Label variants minus retraining with online poses, TraF-Align, three seeds, 95% intervals.

	val. AP@0.7	val. AP@0.5	held-out AP@0.7	held-out AP@0.5
Interpolated labels (selected) − online-pose retraining	+0.81 [0.18, 1.62]	+1.17 [0.58, 1.87]	+3.15 [0.23, 4.81]	+1.91 [−0.67, 3.05]
Interpolated and bound-corrected labels − online-pose retraining	+1.08 [0.37, 1.94]	+1.40 [0.81, 2.10]	+3.74 [0.55, 5.38]	+2.13 [−0.02, 3.12]

Selection criterion fixed in advance. Before evaluating the validation split we required, for the selected variant, that the 95% intervals of the difference to retraining with the online pose lie above zero for both AP@0.7 and AP@0.5, and that the difference to CoMET without retraining be at least +1.0 AP@0.5 with an interval above zero and no worse than −0.30 AP@0.7. All five conditions were met.

Training budget. Table 21 retrains the second-tier network for 2,000, 8,000 and 20,000 updates with otherwise identical settings. The calibration split selects 2,000 updates, and longer training lowers AP@0.7 on every split, also for retraining with the online pose. A plausible reason is that the released weights were already trained on the training sequences, so additional updates over-fit the fit split.

Table 21: Training budget of the second tier (TraF-Align, three seeds, mean over the four latency budgets). Calibration AP@0.7 is the selection criterion; the last column is retraining with the online pose for the same number of updates.

	calibration	validation		held-out	validation, online pose
Updates	AP@0.7	AP@0.7	AP@0.5	AP@0.7	AP@0.7
2,000 (selected)	67.91	66.49	81.44	71.45	65.68
8,000	67.42	64.68	79.31	69.92	64.12
20,000	63.64	60.55	75.45	66.01	60.10

Table 22: Second tier under the auxiliary references: second tier minus retraining with online poses (TraF-Align, 95% intervals).

Reference	Difference	val. AP@0.7	val. AP@0.5	held-out AP@0.7	held-out AP@0.5
interpolated	second tier − online-pose retraining	+1.19 [0.54, 2.00]	+1.79 [1.11, 2.57]	+3.43 [0.90, 4.92]	+2.40 [0.46, 3.26]
label-median	second tier − online-pose retraining	+1.27 [0.56, 2.07]	+2.29 [1.25, 3.72]	+4.00 [1.26, 5.66]	+3.25 [1.52, 4.15]

F Criteria fixed in advance for the two additions

Lock acquisition. All parameters of Section 4.4 were fixed before any result was computed. The step was to be adopted only if (i) without injected error, the mean change of validation AP@0.7 over the five detectors was within ±0.05 points and no detector lost more than 0.10, and likewise on the held-out split; and (ii) with an injected constant error of 3 and 5 m on the validation split, for both TraF-Align detectors, it retained at least 90% of the AP@0.7 of CoMET without injected error and was better than CoMET under the same error with a paired interval above zero. Both conditions were met (Section 6.10); the response to drift and steps was reported without a criterion.

Two-tier combination. The four settings of Section 4.6 were compared on the calibration split only, where IoU 0.3 without isotonic calibration scored highest (69.44 AP@0.7, against 68.94–69.30 for the other three). The combination was to be adopted only if, on the validation split, its AP@0.7 exceeded the better of the two tiers by at least 0.50 points with a paired interval above zero and its AP@0.5 was at most 0.20 points below the second tier, and if its held-out AP@0.7 exceeded both tiers. All conditions were met (Table 13).

G Reproducibility

The repository contains (i) the stand-alone estimator, the lock-acquisition step, the two-tier combination and the pose-error injection, with unit tests and scripts that compare them with the reference per-message states and detections; (ii) a snapshot of the research code that produced every reported result, with a SHA-256 manifest; (iii) every table of this paper in machine-readable

form with the source of each number; and (iv) the scripts that generate the tables and figures of this manuscript from those files. No dataset content, per-frame annotation-derived data or model weights are redistributed.